

Nishant Chourasia

M.Tech CSE | IIT Jodhpur

SYSTEMS / BACKEND / COMPUTER ARCHITECTURE

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PROFILE

M.Tech CSE student at IIT Jodhpur focused on **systems programming, computer architecture, and backend engineering**. Experience in **C++/Python**, concurrent storage, transactional APIs, and performance evaluation; currently researching **LLC cache compression using gem5**.

GATE 2025 / CS

AIR 1,197

Score **706** | Marks **64.16/100**

UGC-NET / JUNE 2024

Qualified

Computer Science & Applications

Ph.D. admission only

EDUCATION

DEGREE / INSTITUTION	CGPA / %	YEARS
M.Tech (CSE) — IIT Jodhpur	7.25	2025–Present
B.Tech (CSE) — Astral Institute Indore (RGPV Bhopal)	7.93	2019–2023
Class 12 — JNV Khandwa M.P. (CBSE)	78.6%	2018–19
Class 10 — JNV Khandwa M.P. (CBSE)	8.6	2016–17

TECHNICAL EXPERTISE

Languages C++, C, Python, SQL, JavaScript/TypeScript

Systems / Architecture gem5, ChampSim, Cache Compression, Hardware Prefetching, Concurrency, Multithreading, Network Programming, epoll, Linux

Backend / Databases FastAPI, PostgreSQL, Redis, REST APIs, Transactions, Authentication, Session Management, Concurrency Control

SYSTEMS & ARCHITECTURE PROJECTS

Cache Compression in Memory Hierarchies

Jan 2026 – Present

M.Tech Thesis, IIT Jodhpur

- Investigating **LLC cache compression** in **gem5** to improve effective cache capacity; evaluating compression effectiveness, cache misses, and system-level performance across benchmark workloads.
- Developing reproducible experiments in **C++/Python** on **Linux**; analyzing write behavior and architectural metrics to compare cache behavior and the performance impact of compression.

SwiftKV

2023

C++ Systems Project

- Engineered a single-node key-value store in **C++20** with **epoll-based event loops**, **64 concurrent storage shards**, persistence, and **RESP-compatible request handling**.
- Measured **1.63M operations/sec** peak throughput under controlled workloads; evaluated scalability, latency, persistence overhead, and connection behavior.
- Validated reliability with **212 C++ test cases**, **AddressSanitizer**, **UBSan**, and **TSan**, crash-recovery checks, and concurrency testing.

ENGINEERING PROJECTS

CampusFlow

2025

Backend / Software Engineering Project

- Built a **FastAPI/PostgreSQL** backend for authentication, course enrollment, assignments, submissions, grading, attendance, and notifications.
- Designed transactional workflows using **FOR UPDATE, SKIP LOCKED, and ON CONFLICT**, partial indexes, and constraints; implemented database-level authorization checks for access control.
- Validated correctness with **284 automated tests** covering enrollment races, seat limits, authorization, and HTTP-level concurrent requests across backend endpoints.

ResumeLens

2025

Full-Stack / AI-Backend Project

- Developed a resume-to-job matching application using **FastAPI, PostgreSQL, and React**, with **deterministic scoring** and optional LLM feedback.
- Implemented authentication, secure **session management**, authorization, ownership checks, **user isolation**, and rate limiting; validated workflows with automated integration and security tests.

SmartLab

Sep – Nov 2025

Machine Learning Project

- Built a reproducible ML pipeline over **241,600 dataset rows**, covering **feature engineering**, model selection, validation, and comparative performance analysis.
- Tuned **HistGradientBoosting** through hyperparameter search, reaching **0.0205 test MAE(log)**; measured a **23.9% feature-engineering gain** in the ML experimentation pipeline.

Hardware Prefetcher Design and Evaluation

Nov – Dec 2025

M.Tech Computer Architecture Project

- Designed and evaluated a hardware prefetching scheme in **ChampSim**, analyzing cache misses, memory behavior, and processor performance across benchmark traces.
- Implemented experimental modifications in **C++** and compared performance using simulator-generated architectural metrics.

TOOLS, METHODS & CS FOUNDATIONS

Tools / Frontend Git/GitHub, Docker, CMake, React, Vite, Testing, AddressSanitizer, UBSan, TSan**ML / Evaluation** Scikit-learn, Feature Engineering, Model Selection, Hyperparameter Tuning, Model Evaluation**CS Foundations** Data Structures & Algorithms, Operating Systems, Advanced Computer Architecture, DBMS, Computer Networks, Theory of Computation, Cyber Security, Blockchain Technology

TEACHING & ACADEMIC RESPONSIBILITY

IIT Jodhpur

Teaching Assistant — Computer Organization & Architecture

2025–Present

Teaching Assistant — Principles of Computer Systems

2025

Lab Instructor — Computer Organization & Architecture Lab (CSL3130)

2025

- Teaching assistant responsibilities included tutorials, assignment evaluation, clarification of systems concepts, and guidance in computer architecture laboratory work.

ACHIEVEMENTS & PARTICIPATION

Runner-Up — CodeAdept 8.0, UIT-RGPV Bhopal

Nov 2024

Team Runner-Up — College Technical Competition (B.Tech)

2023

Adobe University Hackathon — Certificate

Aug 2026